PharmAchieve

INFECTIOUS DISEASE DRUG CHARTS

Qualifying Exam Preparatory Course

LEGEND

Amp	Ampicillin
AOM	Acute Otitis Media
BCG	Bacillus Calmette–Guérin
B-lactams	Beta-lactams
(CA)-MRSA	(Community Acquired)-Methicillin-Resistant Staphylococcus Aureus
CAP	Community-Acquired Pneumonia
Cephs	Cephalosporins
C/I	Contraindication
CNS	Central Nervous System
CrCl	Creatinine Clearance
CYP	Cytochrome P450
DNA	Deoxyribonucleic Acid
ESBLs	Extended-spectrum Beta-lactams
FQ	Fluoroquinolones
G+ve/G-ve	Gram Positive/Negative
G6PD	Glucose-6-phosphate Dehydrogenase
GABHS	Group A Beta-hemolytic Streptococcus
Gen	Generation
GI	Gastrointestinal
HA	Headache
HIV	Human Immunodeficiency Virus
Hr	Hour
IM	Intramuscular
INR	International Normalized Ratio
IV	Intravenous
LFTs	Liver Function Tests

MAC	Mycobacterium Avium Complex
MAOI	Monoamine Oxidase Inhibitor
MOA	Mechanism of Action
Moxi	Moxifloxacin
MSSA	Methicillin-Susceptible Staphylococcus Aureus
MTX	Methotrexate
N/V/D	Nausea/Vomiting/Diarrhea
PBP	Penicillin Binding Protein
PCP	Pneumocystis Pneumonia
PEcK	*Proteus spp, E. coli, Klebsiella spp*
Pen	Penicillin
PID	Pelvic Inflammatory Disease
PO	By Mouth
PRSP	Penicillinase-Resistant Synthetic Penicillin
RNA	Ribonucleic Acid
Rxn	Reaction
SMX/TMP	Sulfamethoxazole/Trimethoprim
SPACE	*Serratia, Pseudomonas, Acinetobacter, Citrobacter, Enterobacter*
SPICE	*Serratia, Providencia spp, Indole-positive Proteus spp, Citrobacter, Enterobacter*
Spp	Species
SSRI	Selective Serotonin Reuptake Inhibitor
Strep	Streptococcus Species
UTI	Urinary Tract Infection
Vanco	Vancomycin
VRE	Vancomycin-Resistant Enterococcus

FOCUSED OVERVIEW OF DRUG INDICATIONS

Indications	Bacteria	Drug/classes	
		1st line therapy	Alternative therapy
CAP	*S. Pneumoniae* *C. Pneumoniae* *M. Pneumoniae* *H. Influenzae*	Macrolides Doxycycline B-lactam	Respiratory fluoroquinolones Amoxicillin-Clavulanic Acid
AOM	*S. Pneumoniae* *H. Influenzae* *M. Catarrhalis*	Amoxicillin Amoxicillin-Clavulanic Acid	Cefuroxime Ceftriaxone Clindamycin
Pharyngitis	*GABHS*	Penicillin Amoxicillin	Cephalexin Clindamycin Macrolides
Sinusitis	*S. Pneumoniae* *H. Influenzae* *M. Catarrhalis*	Amoxicillin	Amoxicillin-clavulanic acid SMX-TMP Macrolides

FOCUSED OVERVIEW OF DRUG INDICATIONS

Indications	Bacteria	Drug/classes	
		1st line therapy	Alternative therapy
Furuncle/Carbuncle	*S. aureus*	Cephalexin Cloxacillin	
Cellulitis	*S. aureus* *Strep. spp*	Cephalexin Cloxacillin Clindamycin	SMX-TMP (MRSA)
Uncomplicated UTI	*E. coli*	Nitrofurantoin SMX-TMP Fosfomycin	FQ Beta-lactams
Intra-abdominal infection (peritonitis)	*E. coli* *B. fragilis* *Streptococcus*	Moxifloxacin Ertapenem 3rd generation cephalosporin	Metronidazole + FQ
C. Difficile infection	*C. difficile*	Vancomycin or fidaxomicin	Pseudomembranous colitis: Vancomycin Severe: IV metronidazole + vancomycin
Meningitis (2-50 years old)	*S. pneumoniae* *N. meningitidis* *H. influenzae*	Cefotaxime / ceftriaxone Add Vancomycin for S. pneumo	Meropenem FQ

BETA-LACTAMS

Class	Drug	Key antimicrobial spectrum
Natural penicillin	Penicillin V	• *Streptococcus*, oral anaerobes
Penicillinase-resistant penicillin	Cloxacillin	• *Streptococcus* • DOC: *MSSA*
Aminopenicillin	Ampicillin, Amoxicillin	• Same as Pen V + *PEc, Enterococcus*
Beta-lactam-beta-lactamase inhibitor	Piperacillin-Tazobactam, Amoxicillin-clavulanic acid	• Same as Aminopen + *MSSA, H. influenzae, M. catarrhalis, Pseudomonas, S. aureus, Klebsiella*
1st generation cephalosporin	Cefazolin, Cephalexin	• *MSSA, PEcK, Streptococci*
2nd generation cephalosporin	Cefuroxime, Cefaclor, Cefoxitin, Cefprozil	• Same as 1st plus *BL-H.influ, m.cat, e.coli, klebsiella, anaerobes* (Cefoxitin)
3rd generation cephalosporin	Cefixime, Cefotaxime, Ceftriaxone, Ceftazidime*	• Better Gram-ve than 2nd generation • Same as 1st plus *n.gon*, oral anaerobe • * DOC: *Pseudomonas*
4th generation cephalosporin	Cefepime	• G+ve and g-ve, *Pseudomonas*, limited anaerobic coverage
Carbapenems	Ertapenem, Imipenem, Meropenem	• *E.coli, Klebsiella, MSSA, pseudomonas, B.fragilis,* anaerobes • DOC: *E. faecalis* (not w. Ertapenem)

NATURAL PENICILLIN

Drug/class	MOA	Safety	Mechanism of resistance
Penicillins (V, G) Bactericidal	All beta-lactams are cell wall active agents that bind to PBP and inhibit cell wall synthesis	• Hypersensitivity (1-10%) • Seizure at high doses IV	• Inactivation with beta-lactamase • Modification of PBP

Clinical pearls

- *NO longer active against S. aureus and N. gonorrhoeae*
- Renal dosing is required
- Oral penicillin has erratic absorption (rarely used)
- Prolonged effects seen with preparation containing benzathine as "depot" (IM)

Coverage

- Aerobic gram-positive cocci: *Beta Hemolytic Streptococci, Enterococci (E.faecalis)* and *S. viridans* (if sensitive)
- *Neisseria meningitidis* (if sensitive), *Treponema palladium* (syphilis)
- Oral anaerobes
- Gut anaerobes: Clostridium species *(C. tetani, C. perfringens)*

PENICILLINASE-RESISTANT PENICILLIN

Drug/class	MOA	Safety	Mechanism of resistance
Cloxacillin *Semi-synthetic penicillin*	Same as all beta lactams • Bactericidal • Resistant to beta-lactamase	• N/V • Rash, anaphylaxis	• Same as all beta lactams

Clinical pearls
- IV preferred
- Oral cloxacillin is poorly absorbed – food ↓ absorption
- Administer on empty stomach (1 hr before or 2 hrs after meals)

Coverage
- *Staphylococcus (not MRSA)* → *drug of choice*
- *Streptococci* (inferior to other penicillins), *H.influenzae*
- ↓ activity against oral anaerobes
- *NO activity against enterococci*

AMINOPENICILLINS AND BL-BLI

Drug/class	MOA	Safety	Mechanism of resistance
Amoxicillin/ Ampicillin or Amox-clav Bactericidal	Same as all beta lactams • Amox: amino group added to penicillin • Clav: binds to beta-lactamases	• Rash • N/V/D (note: amox-clav has MORE diarrhea than amox alone)	• Same as all beta lactams • Amox/amp no anti-beta-lactamase

Coverage for amox/ampicillin

- Everything penicillin covers PLUS some:
  - G-ve bacilli including *H. influenzae* (beta-lactamase negative), *Proteus mirabillis, E. coli, strepptococci*
- *Listeria monocytogenes, E. faecalis*
- Oral anaerobes (no gut anaerobes)
- *Does NOT cover Kleb, S. aureus, atypicals*

Coverage for amox-clav

- Everything amoxicillin covers PLUS:
  - BL producing *H. influenza, PEcK, M. catarrhalis*
  - *E.faecalis*, Staphylococcus (MSSA), Streptococcus
  - Some coverage for gut anaerobes (*B. fragilis*)

BL-BLI: PIPERACILLIN-TAZOBACTAM

Drug/class	MOA	Safety	Mechanism of resistance
Piperacillin-tazobactam	Same as all beta lactams Bactericidal • Tazobactam is β-lactamase inhibitor	• N/V • Rash, anaphylaxis	• Same as all beta lactams

Clinical pearls

- Do not use for definitive therapy for SPACE/SPICE organisms even though beta-lactam inhibitor added

Coverage

- Everything ampicillin covers PLUS extends spectrum to include beta-lactam producing organisms: PEcK, *H. influenzae*, gut & oral anaerobes, *M. catarrhalis*
- Covers *Pseudomonas*, if susceptible

1ST GEN CEPHALOSPORINS

1st 2nd 3rd 4th

Gram(+) Generation of Cephs Gram (-)

Drug/class	MOA	Safety	Mechanism of resistance
Cefazolin (IV), Cephalexin (PO), Cefadroxil (PO) • Bactericidal	Same as all beta lactams	• N/V/D, *C. difficile* diarrhea (variable) • Hypersensitivity (cross-reaction w/pen <5%)	• Same as all beta lactams

Coverage

- G+ve cocci, MSSA, *Streptococci* → Very good
- *S. epidermidis*
- Oral anaerobes
- *NO activity against: H. influenza(+/-), M. catarrhalis, anerobes, SPACE, Listeria, MRSA, MRSE, atypicals*

2ND GEN CEPHALOSPORINS

1^{st} 2^{nd} 3^{rd} 4^{th}

Gram(+) Generation of Cephs Gram (-)

Drug/class	MOA	Safety	Mechanism of resistance
Cefuroxime (IV, PO), Cefprozil (PO), Cefoxitin (IV)	Same as all beta lactams	• Same as 1^{st} gen	• Same as all beta lactams

Coverage

- Cefuroxime/cefaclor: *H. influenzae* (including beta-lactam producing), *M. catarrhalis*, oral anaerobes, Streptococci
- Cefoxitin: Covers PEcK plus gut anaerobes (B. fragilis)
- *NO activity against: MRSA/MRSE, Enterococcus spp, Listeria spp, anaerobes (except Cefoxitin, atypicals)*
- Cefaclor, cefuroxime: upper respiratory tract infection (RTI)
- Cefoxitin: uncomplicated intra-abdominal infections, pelvic inflammatory disease, & surgical procedures, infections due to G-ve plus anaerobes

3RD GEN CEPHALOSPORINS

1st 2nd 3rd 4th
Gram(+) Generation of Cephs Gram (-)

Drug/class	MOA	Safety	Mechanism of resistance
Cefotaxime (IV), Ceftriaxone (IV), Ceftazidime (IV), Cefixime (PO)	Same as all beta lactams	• Same as 1st gen	• Same as all beta lactams

Clinical pearls

- Better CNS penetration compared to other classes of cephalosporins
- Use ceftazidime if concern for Pseudomonas infection and NOT if concern for G+ve organism
- NOT recommended in “SPICE” infections

Coverage

Multiple indications including pneumonia, UTI/pyelonephritis, meningitis, intra-abdominal infection in combination with metronidazole (for gut anaerobes)

- PEcK ↑, *H. influenzae*, *M. catarrhalis*
- Ceftriaxone: good Strep coverage including penicillin-resistant *S. pneumoniae*.
- Ceftazidime: same G-ve coverage compared to ceftriaxone but no G+ve activity,
- Cefixime: good activity for *N. gonorrhoeae*
- *NO activity against: MRSA/MRSE, Enterococci, Listeria spp, gut anaerobes, Legionella*
- *Cefixime has poor activity against streptococcus or staphlococcus*

4TH GEN CEPHALOSPORINS

1st 2nd 3rd 4th
Gram(+) Generation of Cephs Gram (-)

Drug/class	MOA	Safety	Mechanism of resistance
Cefepime (IV)	Same as all beta lactams	• Same as 1st gen	• Same as all beta lactams

Clinical pearls
- Good for mixed infections (G-ve, G+ve, Pseudomonas)
- Useful for difficult to treat organism such as SPICE/SPACE

Coverage
- *Covers everything ceftriaxone covers*
- Improved G+ve (Staphylococcus/Streptococcus) coverage, good G-ve coverage (including Pseudomonas)
- Retains activity for SPACE organisms. Eg. has activity against broad spectrum beta-lactamase producing G-ve (Amp C beta-lactamase, and maybe Class A ESBLs)
- *NO activity against: gut anaerobes*

CARBAPENEMS

Drug/class	MOA	Safety	Mechanism of resistance
Imipenem (IV), Meropenem (IV), Ertapenem (IV)	Same as all beta lactams	• Imipenem, meropenem associated with ↓seizures threshold (at high doses) • *C. difficile*	• Same as all beta lactams

Clinical pearls

- Useful for difficult to treat organism with multi-resistance such as SPACE/SPICE, ESBLs
- Good for mixed infections (G-ve, G+ve, *Pseudomonas*)
- Do NOT use ertapenem for *Pseudomonas spp* or *Acinetobacter spp* (no/unreliable coverage)

Coverage

- G-ve (PEcK,SPACE/SPICE) , G+ve (*Streptococcus, Staphylococcus*), anaerobes (gut/oral)
- Some activity against *Enterococcus faecalis* (but ↑ failure rate)
- *NO activity against: MRSA, atypical organism (Mycoplasma spp, Chlamydia spp), C. difficile*

PROTEIN SYNTHESIS INHIBITORS

Macrolides

- Erythromycin, azithromycin, clarithromycin
- Active against: *S. pneumoniae, Hemophilus influenzae, Moraxella catarrhalis, Chlamydophila, Legionella, Mycoplasma pneumoniae*

Lincosamides

- Clindamycin, lincomycin
- Active against: *Staphylococci, Streptococcus viridans, Strep. pyogenes, Strep. pneumoniae, MRSA* (Clindamycin)

Aminoglycosides

- Streptomycin, tobramycin, gentamycin, neomycin
- Active against: *Pseudomonas, Enterobacter, Mycobacterium tuberculosis, SPACE organisms*

Tetracyclines

- Doxycycline, minocycline, tigecycline
- Active against: some MRSA, *B. anthracis, Mycoplasma, Chlamydophila sp, Helicobacter pylori*

MACROLIDES

Drug/class	MOA	Safety	D/I
Macrolide: Erythromycin (IV/PO), Clarithromycin (PO), Azithromycin (PO,IV) • Bacteriostatic	Inhibits protein synthesis (prevents translocation of t-RNA)	• GI (N/V, abdominal cramping), cholestatic jaundice, metallic taste (clarithromycin)	• CYP3A4 enzyme (erythromycin = most interacting; azithromycin = least interacting)
Clinical pearls • GI effects are more common with erythromycin • Erythromycin estolate is contraindicated in pregnancy due to irreversible hepatotoxic effects. Other erythromycin salts may be used instead.			
Coverage • G+ve including Streptococcus spp. Weak activity against Staphylococcus spp • H. influenzae, M. Catarrhalis (except for erythromycin) • Excellent atypical coverage (Mycoplasma, Legionella, Chlamydia) • N. gonorrhoeae, B. pertussis, Campylobacter, Lyme, H. pylori, MAC • *NO activity against: MRSA, Enterococcus spp, gut anaerobes, most G-ve*			

CLINDAMYCIN

Drug/class	MOA	Safety
Lincosamide: clindamycin	Inhibits protein synthesis (50S) and toxic synthesis • Bacteriostatic	• N/V/D • Hemolytic reactions in pts with G6PD deficiency • Pseudomembranous colitis (*C.difficile* diarrhea)

Clinical pearls

- *Take with food*
- Useful for pts allergic to pen requiring G+ve & anaerobic coverage, some MRSA coverage
- Also useful for PID, bacterial vaginosis, toxic shock syndrome (if combined with beta-lactam)
- Used topically for acne

Coverage

- Oral anaerobes (*Peptostreptococcus, Fusobacterium, Clostridia*), gut anaerobes (*Bacteroides fragilis, Clostridium spp*), aerobic G+ve *Staphylococcus spp, Streptococci spp,* Protozoa
- *NO activity against: Enterococcus, most G-ve*

AMINOGLYCOSIDES

Drug/class	MOA	Safety	Monitoring
Gentamicin (IV) Tobramycin (IV) Amikacin (IV) Streptomycin (IV) • Bactericidal	Disruption of bacterial protein synthesis (30S)	• Ototoxicity (esp. if high dose, long-term use), • Nephrotoxicity (esp. high troughs) • Post-antibiotic effect seen with G-ve organisms	• Monitor pre (trough) & post (peak) level with q8-12 hour dosing • Do not need to monitor for once-daily dosing

Clinical pearls

- Useful for difficult to treat G-ve organism such as *Pseudomonas &* urosepsis, bacteremia, in combination with other drugs for multi-resistant organisms
- Gentamicin used for synergism with pen or vancomycin for MSSA/MRSA and Enterococcus
- Do NOT use in pregnancy

Coverage

- G-ve (SPACE/SPICE organisms): Gentamicin has best activity for *Serratia* & *Klebsiella*, Tobramycin has best activity for *Pseudomonas*, Amikacin usually has lower rates of resistance *NO activity against: G+ve, atypical organism, anaerobes*

TETRACYCLINES

Drug/class	MOA	Safety	D/I
Tetracycline (PO) Doxycycline (PO) Minocycline (PO)	Inhibits protein synthesis • Bacteriostatic	• Photosensitivity • Discolouration of teeth, use in those ≥8-13 years old depending on the agent (doxycycline has lower risk) • N/V	• Take 2 hours before/after dairy/multivalent cations

Clinical pearls

- *Streptococcus, staphylococcus, n. gonorrhoeae* resistance ↑
- Do NOT use in pregnancy
- *Take on empty stomach (1 hr before/2 hr after); EXCEPT minocycline is taken with or without food*

Coverage

- G+ve (S. aureus, Streptococci including S. pneumoniae) & G-ve (H. influenzae, M. catarrhalis)
- Atypical coverage (Mycoplasma, Chlamydia, Legionella) excellent
- Lyme disease, Rickettsia
- NO activity against: anaerobes and hard to kill G-ve

TIGECYCLINE

Drug/class	MOA	Safety	D/I
Tigecycline (IV)	Inhibits protein synthesis • Bacteriostatic	• ↑LFTs, myopathy • Anemia, thrombocytopenia • Jaundice, bloating • N/V/D/HA • Taste disturbance, phlebitis	• ↑ warfarin but no INR change • ↓ COC efficacy

Clinical pearls

- Static drug, low serum/bloodstream concentrations
- Useful for MRSA when vancomycin, linezolid, daptomycin is not an option
- ↑ mortality compared to other agents
- Do NOT use in pregnancy

Coverage

- G+ve (*S. aureus* including MRSA, Streptococci, Enterococcus including VRE)
- G-ve (*H. influenzae, M. catarrhalis, E.coli*)
- Atypical coverage (Mycoplasma, Chlamydia, Legionella) excellent and gut anaerobes
- *NO activity against:* Proteus, Providencia, and Pseudomonas

DNA SYNTHESIS INHIBITORS

Fluoroquinolones

- Ciprofloxacin, moxifloxacin, levofloxacin
- Active against: *PEcK, Hemophilus influenzae, Moraxella Catarrhalis,* SPACE/SPIC, *Neisseria gonorrhoeae*

Metronidazole

- Metronidazole
- Active against: *Peptostreptococcus spp, Fusobacterium spp, Bacteroides fragilis, Clostridium spp*

FLUOROQUINOLONES

Drug/class	MOA	Safety	D/I
Norfloxacin (PO), Ciprofloxacin (IV, PO), Levofloxacin (IV, PO), Moxifloxacin (IV, PO) • Bactericidal Ofloxacin is only available as eye drops	• Inhibits DNA replication (DNA gyrase inhibition)	• Tendonitis (Achilles tendon rupture), diarrhea, *C.difficile*, QTc prolongation	• CYPP450 substrate (not moxi) • Moxi is the only FQ that does NOT require renal dose adjustment • Take 2 hrs before/after dairy/multivalent cations

COVERAGE

- G-ve (PEcK, *H. influenza*, *M. catarrhalis*, SPACE/SPICE, *N. gonorrhoeae*, *Salmonella*, *Shigella*)
- Atypical organism (*Mycoplasma pneumonia*, *Chlamydia pneumonia*, *Legionella pneumophilia*)
- G+ve streptococci including penicillin-resistant strains (only for respiratory FQ: levofloxacin, moxifloxacin)
- Moxifloxacin also covers anaerobes (only quinolone with anaerobic coverage)
- Do NOT use Moxifloxacin for UTI because does not penetrate urine adequately
- *NO activity against: MRSA, Pseudomonas spp (except ciprofloxacin/levofloxacin), G+ve (unless respiratory FQ); variable activity against Enterococcus*
- Do NOT use in pregnancy

METRONIDAZOLE

Drug/class	MOA	Safety	D/I
Metronidazole	• Inhibits nucleic acid synthesis • Bactericidal	• N/D/GI upset • ↓ appetite • Neuropathy • Taste disturbance	• Mild sulfiram-like rxn with alcohol (flushing, HA, N/V) • Avoid BCG vaccine • Inhibits 2C9 - use lithium and warfarin with caution – monitor as needed

Clinical pearls

- Useful for mild *C.difficile* (PO > IV); IV used as adjunct with vancomycin for severe cases
- Can be used for protozoa, *H. pylori* (2^{nd} line)

Coverage

- Oral anaerobes (*Peptostreptococcus spp, Fusobacterium spp*) and gut anaerobes (*Bacteroides fragilis, Clostridium spp*), other protozoa & *H. pylori*
- *NO activity against: G+ve, most G-ve, atypical*
- Do NOT use in 1st trimester of pregnancy or breastfeeding

FOLATE SYNTHESIS INHIBITOR

Sulfamethoxazole-trimethoprim

TRIMETHOPRIM/SULFAMETHOXAZOLE

Drug/class	MOA	Safety	D/I
Trimethoprim/ Sulfamethoxazole	Prevents folic acid synthesis • Bacteriostatic	• Hyponatremia • Anemia, hemolysis • Hyperkalemia • N/V/D/rash • Photosensitivity	• Folic acid may be co-administered without affecting effect • Caution with MTX, Phenytoin; Avoid Warfarin, Cyclosporin

Clinical pearl
- Drug of Choice: Pneumocystis jirovecii pneumonia, Nocardia spp infection, S. maltophilia, CA-MRSA, and uncomplicated cystitis
- Useful for pts with pen allergy for Listeria infections

Coverage
- G+ve: *Staphylococcus, streptococcus,* (S. *aureus* possibly including MRSA), Listeria, Nocardia
- G-ve: PEcK, *H. influenzae, M. catarrhalis,* SPACE/SPICE but not Pseudomonas
- *Stenotrophomonas maltophilia,* PCP and toxoplasmosis (in HIV patients)
- *NO activity against: Enterococci, atypicals, anaerobes, Pseudomonas. Poor against S. pneumo*

OTHERS

- Vancomycin
- Daptomycin
- Fosfomycin
- Linezolid
- Nitrofurantoin

VANCOMYCIN

Drug/class	MOA	Safety	Monitoring
Vancomycin	Inhibits cell wall synthesis in G+ve bacteria • Bactericidal	• Nephrotoxicity, Vancomycin Infusion Syndrome, (↓ infusion rate), thrombophlebitis, ototoxicity • Stomatitis with oral use, hypotension with rapid IV infusion	• Monitor pre (trough) levels with the 3rd/4th IV dose • PO vancomycin doses are not absorbed since it is a large molecule, but it is ideal for C. *difficile*

Clinical pearls

- Issue with resistance from overuse
- IV form given as PO for *C. difficile* in hospital due to increased cost of PO vancomycin
- Vancomycin dose needs to be adjusted for obesity, burns, critical illness, renal dysfunction, pregnancy and cystic fibrosis
- Actual body weight should be used to dose vancomycin loading and maintenance doses

Coverage

- G+ve (*S. aureus*, MRSA, *S. epidermidis*, *Streptococcus spp*, *Enterococcus spp*, *Listeria*)
- *C. difficile* when used orally
- *No activity against: G-ve (except Neisseria), atypical, anaerobes*

DAPTOMYCIN

Drug/class	MOA	Safety	D/I
Daptomycin	Inhibits cell wall synthesis • Bactericidal	• Myopathy, ↑LFTs, ↑INR • Neuropathy • Hypersensitivity, ↓K, Mg • Eosinophillic pneumonia	• Statins may ↑ myopathy and precipitate AKI

Clinical pearls

- Do NOT use for pneumonia - drug is inactivated by lung surfactant
- Commonly used for SSTI by G+ve
- Needs to be adjusted for renal function
- Useful for drug resistant Staphylococcus/Enterococcus; can use for pts failing vancomycin and MRSA, MRSE
- Prolonged use may ↑ resistance especially in VRE and MRSE

Coverage

- G+ve (*S. aureus*, MRSA, *Streptococcus spp*, *Enterococcus spp* including VRE (both *faecium* and *faecalis*)
- *NO activity against: G-ve, atypical, anaerobes*

LINEZOLID

Drug/class	MOA	Safety	D/i
Linezolid	Blocks initiation of protein production • Bacteriostatic • with some bacteriocidal activty against strep., staph.)	• Bone marrow suppresion, ↑LFTs • Neuropathy • Lactic acidosis • N/V/D/HA	• Inhibits MAOI – can cause serotonin syndrome if used with SSRIs • Enhances sympathomimetic effect of agents such as pseudoephedrine

Clinical pearls

- Potential resistance with overuse; expensive, used as a reserve antibiotic
- Useful for MRSA, VRE, MRSE (serious G+ve infection), SSTI's
- Useful for pts with renal dysfunction who cannot take vancomycin for MRSA, PRSP, VRE

Coverage

- G+ve (*S. aureus*, MRSA, Streptococcus including penicillin resistant *S. pneumoniae* (PRSP), *Enterococcus spp* including VRE (both *faecium* and *faecalis*)
- *NO activity against: G-ve, atypical, anaerobes*

NITROFURANTOIN

Drug/class	MOA	Safety	D/I
Nitrofurantoin	DNA & Cell wall synthesis inhibitor • Bactericidal	• Headache • Flatulence • Nausea & ↓ appetite • C/I if CrCl < 60 mL/min	• Magnesium trisilicate (antacids) • Norfloxacin • Spironolactone • Probenecid

Clinical pearls

- 1st line in uncomplicated UTI
- *Take with food or milk*
- Make cause discolouration of urine (rust yellow or brown)
- Can be used in early stages of pregnancy - CI at term and during labour

Coverage

- *PEcK, S. aureus, S. saprophyticus, S. agalactiae, Enterococcus spp. & faecalis*
- *NO activity against: atypical, anaerobes, Pseudomonas*

FOSFOMYCIN

Drug/class	MOA	Safety	D/I
Fosfomycin	Inhibits bacterial cell wall synthesis • Bactericidal	• Diarrhea • Headache • Nausea • Vaginitis	• Metoclopramide and cisapride slow down absorption
Clinical pearls • *Food delays absorption – administer on an empty stomach* • Pregnancy: No evidence of risk in humans but studies inadequate • Oral bioavailability ~40% • Used as a single dose for uncomplicated UTI's			
Coverage • Gram positive: *Staphylococcus spp., S. aureus (MRSA), Enterococcus spp* • Gram negative: *E. coli, Enterobacter spp., Citrobacter spp., Klebsiella spp., Proteus spp. Providencia spp. P. aeruginosa* • *NO activity against: atypical, anaerobes*			

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CHANGES LOG

March 2025

- Slide 2: Updated legend
- Slide 3: Changed CAP first-line therapy from macrolide +beta-lactam to beta-lactam in accordance to our CAP notes
- Slide 4: Changed meningitis age range from 2-60 to 2-50 yrs of age in accordance to our meningitis notes. Also changed pathogen from E coli to H. infleunza for this age group
- Slide 4: Added peritonitis in brackets next to intra-abdominal infection and changed cefoxitin to '3rd generation cephalosporin' in line with our notes. Cefoxitin is a 2nd generation cephalosporin
- Slide 5,14: Removed doripenem as an example of a carbapenem as it is discontinued
- Slide 16: Removed ketolide telithromycin as it is discontinued
- Slide 22: Removed Gemifloxacin as it is discontinued

April 2025

- Slide 19: Updated age restrictions for tetracyclines

July 2025

- Content reviewed, no changes made

October 2025

- Slide 11: Removed cefaclor as it is cancelled
- Slide 22: Removed ofloxacin as it is cancelled

February 2026

- Slide 22: Added note that ofloxacin is only available as eye drops